

ANUSHA MANDEPUDI

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SUMMARY

I am a **Data Engineer and Analyst** with 4 years of experience in building scalable data engineering solutions involving ETL pipelines, big data processing, and cloud technologies. I am skilled in SQL, Python, R, and distributed systems such as Apache Spark and Hadoop. Experienced in preparing high quality datasets for machine learning models, collaborating with software engineers, and ensuring compliance with industry standards including HIPAA. Passionate about automation, optimizing performance, and delivering reliable analytics pipelines.

SKILLS

Methodologies: SDLC, Agile, Waterfall

Language: Python, R, SQL, Scala, SAP

IDEs: Visual Studio Code, PyCharm, Jupyter Notebook

Packages: NumPy, Pandas, Matplotlib, SciPy, ggplot2, Apache Spark, Sckit-learn, Pyspark

Visualization Tools: Tableau, Power BI

Certifications:

Database Management Badge: Grand Valley State University

Other Skills: JIRA, Confluence, Hadoop,

Google BigQuery, Strategic Planning, Kafka

Cloud Technologies: AWS, Azure

EXPERIENCE

Data Engineer

ABIM | Pennsylvania- Remote | Feb 2025-Present

- Managed ETL workflows using **SSIS** for ingesting, transforming, and loading structured and semi-structured data (CSV, JSON, XML) from **Azure Blob Storage** into **Snowflake Data Warehouse**.
- Executed data transformations and filtering using Apache Spark on Hadoop for high volume student datasets, then staged outputs in Azure Blob Storage to be consumed by SSIS and ADF workflows into Snowflake.
- Developed and optimized **complex SQL queries** for data extraction, transformation, and loading (ETL) within **Snowflake** and **SQL Server** to ensure efficient data workflows.
- Built hybrid **ETL pipelines** using **Azure Data Factory** to orchestrate SSIS packages, manage workflows, and integrate Snowflake, and Blob Storage, with file system watchers and conditional execution logic.
- Enabled **CI/CD** deployment of SSIS packages using **Azure DevOps**, integrated with **Git** repositories and parameterized configurations, monitored execution with Release Pipelines, **Azure Monitor**, and email notifications.
- Built modular SQL-based transformation layers in Snowflake using **DBT** style modeling principles (staging, intermediate, and mart layers) to support analytics workflows and improve code reusability.
- Implemented automated tests for **model validation** using SQL based checks to detect data quality issues (nulls, duplicates, referential mismatches), reducing downstream dashboard errors by **30%**.
- Designed clean and well-structured **data models** in Snowflake to support accurate reporting and analysis of student data, improving data quality and query performance.
- Documented data models with column level metadata, source definitions, and business logic to align with **DBT's documentation standards** and ensure model auditability.
- Collaborated with BI teams to provision curated datasets into **Power BI**, implementing row-level security and building reports visualizing student metrics, maintained lineage docs and data flow diagrams for traceability.

Data Analyst

Grand Valley State University | GR-Onsite | Jan 2023-Dec 2024

- Conducted **statistical analysis** to uncover correlations between parking violations and enforcement effectiveness, providing actionable recommendations.
- Built interactive dashboards in **Tableau** to track parking lot usage, peak hours, and permit trends for strategic planning.
- Enhanced **Tableau** dashboards with dynamic filters, calculated fields, and parameter controls, enabling stakeholders to drill down into parking trends by lot, time, and permit type improving data driven decision making across departments.
- Cleaned and transformed multi source parking data using **Python** (Pandas, NumPy) and managed ingestion through **MySQL** and MongoDB with stored procedures.
- Used **Microsoft Excel** for quick data exploration, pivot table summaries, and validating ETL results against source data.
- Collaborated with university administrators to integrate insights into real-time decision-making, directly influencing **permit distribution** and enforcement policies for the academic year.

- Extracted EHR data from **Cerner Millennium** using CCL scripts and **REST APIs**, including patient demographics, encounters, labs, medication history, and procedures.
- Ingested raw data into Amazon S3 via **AWS DataSync** and automated pipeline triggers using **Lambda** and S3 events.
- Transformed and cleaned data using **Python** in **AWS Glue**, and applied business rules, joins, and ICD/CPT logic using **PL/SQL** in **Amazon RDS** for SQL Server and utilized **DynamoDB** for storing reference data used in transformation logic.
- Processed real time **Medicaid** claims from Cerner using CDT codes, improving classification accuracy by 45%.
- Modeled and loaded curated datasets into **Amazon Redshift** using star schema design; implemented clustering keys, materialized views, and row level security to improve query performance and enforce **HIPAA** compliant access control.
- Designed reusable data marts for population health metrics, care gaps, utilization patterns, and claims based **KPIs** supporting downstream analytics and regulatory reporting.
- Delivered HIPAA compliant workflows by masking **PHI/PII**, implementing audit logs, and validating adherence to security protocols across the entire data pipeline lifecycle.
- Provided production support for critical healthcare analytics workflows, maintaining data accuracy with **HIPAA**.
- Developed predictive models in **AWS SageMaker** to generate chronic disease risk scores; integrated outputs into dashboards via **QuickSight**.
- Assisted in maintaining **CI/CD** pipelines by integrating automated testing and deployment processes using Jenkins and GitHub Actions, helping streamline ETL job releases and reduce manual errors.
- Coordinated with **development and operations** teams to troubleshoot and resolve build or deployment failures during automated pipeline runs.
- Collaborated with the **QA team** to design and implement data validation tests for healthcare ETL pipelines, reducing post-deployment data issues by **25%** and improving overall data reliability.
- Tracked tasks, production issues, and sprint progress using **Jira** as part of an agile workflow supporting data engineering and analytics initiatives.

Projects & Achievements

- **Early Diagnosis of Lung Cancer:** Developed a machine learning model using K-Nearest Neighbors (KNN) and Logistic Regression in Python to predict lung cancer with high accuracy 85%.
- **Cloud-Based Image Analysis and Translation Application:** Developed a cloud-based image translation and analysis application using Cloud Storage, Cloud Functions, and Cloud Vision API with robust security (IAM), enabling advanced image analysis, multimedia support, custom ML models, and real-time collaboration.
- **Shop App (E-commerce website):** Played a key role in developing the Shop App, using React JS for a seamless user experience and Firebase for a secure, high-performance backend, redefining efficiency and innovation in online shopping.
- **Location Prediction on Twitter using Machine Learning Techniques:** Developed algorithms to predict user locations from social media data, built an intuitive evaluation tool, and designed a comprehensive guide for targeted site analysis

EDUCATION

Master's in computer science, Grand Valley State University, Grand Rapids

Jan 2023-Dec2024